TypeScript Assignment – Requirement Document
Order Management System

Objective

Build a TypeScript-based Order Management System to demonstrate understanding of:
- Interfaces & Types
- Classes & Object-Oriented Programming
- Functions with Optional Parameters
- Generics
- Modules (Import/Export)

Project Structure (Mandatory)

Follow this exact structure:

```
src/
 ├── Product.ts
 ├── Order.ts
 ├── OrderService.ts
 ├── utils.ts
 ├── index.ts
```

Do not change file names or structure.

Functional Requirements

1. Interfaces –

Product Interface: **Product.ts**

- id: number
- name: string
- price: number

Order Interface: **Order.ts**
- id: number
- products: Product[]
- discount?: number

2. OrderService Class – OrderService.ts

Implement the following methods:
- createOrder(products: Product[]): Order
  Order ID should start from 1 and increment

- calculateTotal(order: Order): number

- applyDiscount(order: Order, discount?: number): number
  Discount is percentage-based (e.g., 10 means 10%)
  If no discount is provided, return original total

3. Generics

Implement the following generic function in utils.ts:

```
function filterExpensiveProducts<T extends { price: number }>(
  items: T[],
  minPrice: number
): T[]
```

Requirements:
- Return items where price >= minPrice
- Must use generic <T>
- Must use type constraint (extends)

4. Modules

Use proper import and export statements.
Do not write everything in a single file.

5. Entry File

index.ts should:
- Create sample products
- Create an order
- Apply discount
- Call generic function
- Print results using console.log

Validation Requirements

- Product name should contain only alphabets
- Price must be greater than 0
- Discount should not exceed 100

Testability Requirements

- All methods must return values
- Do not use Math.random()
- Do not hardcode outputs
- Follow exact method names and structure

Constraints

- Use TypeScript only
- Do not use external libraries
- Do not modify function names
- Do not modify file name or structure

Evaluation Criteria (Total: 20 Marks)

Validation & Type Safety – 5
Class Logic (OrderService) – 5
Generics Implementation – 5
Functions (Optional Parameter) – 2
Modules & Structure – 2
Code Quality – 1

Deliverables

Submit "src" folder only with the files as given in the screenshot below. Create a folder (Example: TS Assignment) in your google drive. Copy "src" folder and then share the link of "TS Assignment" folder. Here student 1 is for evaluating purpose and you may keep any name for your folder.

